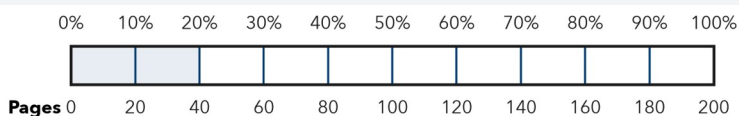


LESSON 4-5 • SESSION 1 • EXPLAIN YOUR THINKING**Find the Whole**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS **part** (*parte*) · **whole** (*entero*) · **double number line** (*doble recta numérica*)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*

Jada has read 40 pages. That is 20% of her book. How many pages are in the whole book?

**HOW I SOLVED IT**

- 1 Match the part.** 40 pages = 20% = 2 parts of 10%.
- 2 Find one part.** 10% = $40 \div 2 = 20$ pages.
- 3 Build the whole.** 100% = 10 parts = $10 \times 20 = 200$ pages.
Check: $0.20 \times 200 = 40$.

I SAY

"The part is **40** pages. It is **20%** of the whole." "So 10% is **20**, and 100% is **200**."

I WRITE

One part of the bar is **20 pages**, so all 10 parts are **200 pages**. The whole is greater than 40 because 20% is less than 100%.

2 • DRAW IT*your turn: label the whole, the part and the percent*

A bike club has saved \$27 for a trip. That is 30% of the money it needs. How much money does the club need in all?

**TRY THESE STEPS** *fill in the blanks as you work*

- \$27 is 30%, which is _____ parts of 10%.
- Find one part: $\$27 \div 3 = \$$ _____.
- Build the whole: $10 \times \$$ _____ = \$_____.

3 • SAY IT*tell a partner*

Partner A: "The part is _____. It is _____% of the whole." Partner B: "So 10% is _____, and 100% is _____." Then switch.

4 • WRITE IT*pick a starter · name the whole · use units***WORD BANK** **the part** · **the whole** · **one part** · **10%** · **greater than** · **in all**

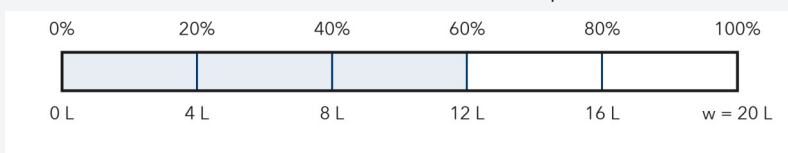
Because I know 10% is _____ because _____ $\div$ _____ = _____.

But Some students think the whole is _____, but the whole must be greater than the part because _____.

So One part of the bar is _____, so all 10 parts are _____.

LESSON 4-5 • SESSION 2 • EXPLAIN YOUR THINKING**Solve for the Whole**

Study the model. Then solve your own problem the same way. Use only the help you need.

WORDS **variable** (*variable*) • **equation** (*ecuación*) • **inverse operation** (*operación inversa*)**1 • SEE A MODEL***read it with a partner***SEE A MODEL** *a problem like yours, solved*A tank has 12 liters of water. That is 60% of the whole tank. Write an equation with w for the whole tank. Solve it.**HOW I SOLVED IT**

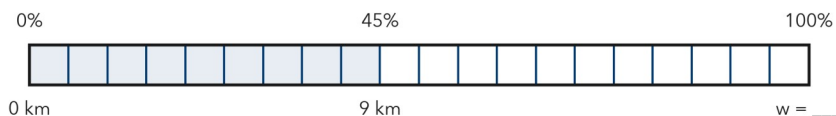
- 1 Write the equation.** percent $\times$ whole = part: $0.60 \times w = 12$.
- 2 Divide.** $w = 12 \div 0.60 = 120 \div 6 = 20$ liters.
- 3 Check.** $0.60 \times 20 = 12$. The whole (20 L) is greater than the part (12 L).

I SAY

"The part is **12 liters** and the percent is **60%**."
 " $0.60 \times w = 12$, so $w = 20$." "Check: $0.60 \times 20 = 12$."

I WRITE

$0.60 \times w = 12$, so $w = 12 \div 0.60 = 20$ liters. I divide because **I know the part and the percent, and I need the whole.**

2 • DRAW IT*your turn: label the whole, the part and the percent*A runner has run 9 km. That is 45% of the whole race. Write an equation with w for the whole race. Solve it.**TRY THESE STEPS** *fill in the blanks as you work*

- 1** Write the percent as a decimal: $45\% =$ _____.
- 2** Equation: _____ $\times w = 9$.
- 3** Divide: $w = 9 \div$ _____ = _____ km. Check by multiplying.

3 • SAY IT*tell a partner*

Partner A: "The part is _____ and the percent is _____." Partner B writes _____ $\times w =$ _____ and solves. Both check: _____ $\times$ _____ = _____.

4 • WRITE IT*pick a starter • name the whole • use units***WORD BANK** **equation** • **variable (w)** • **divide** • **multiply to check** • **part** • **whole****Because** I divide _____ by _____ because _____.**But** Some students multiply _____ by _____, but the whole must be greater than _____.**So** _____ $\times w =$ _____, so $w =$ _____ $\div$ _____ = _____.